4 A curve has parametric equations

$$x = t^3 - t^2 + t - 1 \qquad \text{and} \qquad y = te^t.$$

(a) Show that 1 is the only real value of t for which $x = 0$. [1]

(b) Show that $\dfrac{dy}{dx} = \dfrac{(t+1)e^t}{3t^2 - 2t + 1}$. [3]

(c) Find the Maclaurin's series for y up to and including the term in x^2. [6]